

self-supervised representation learning

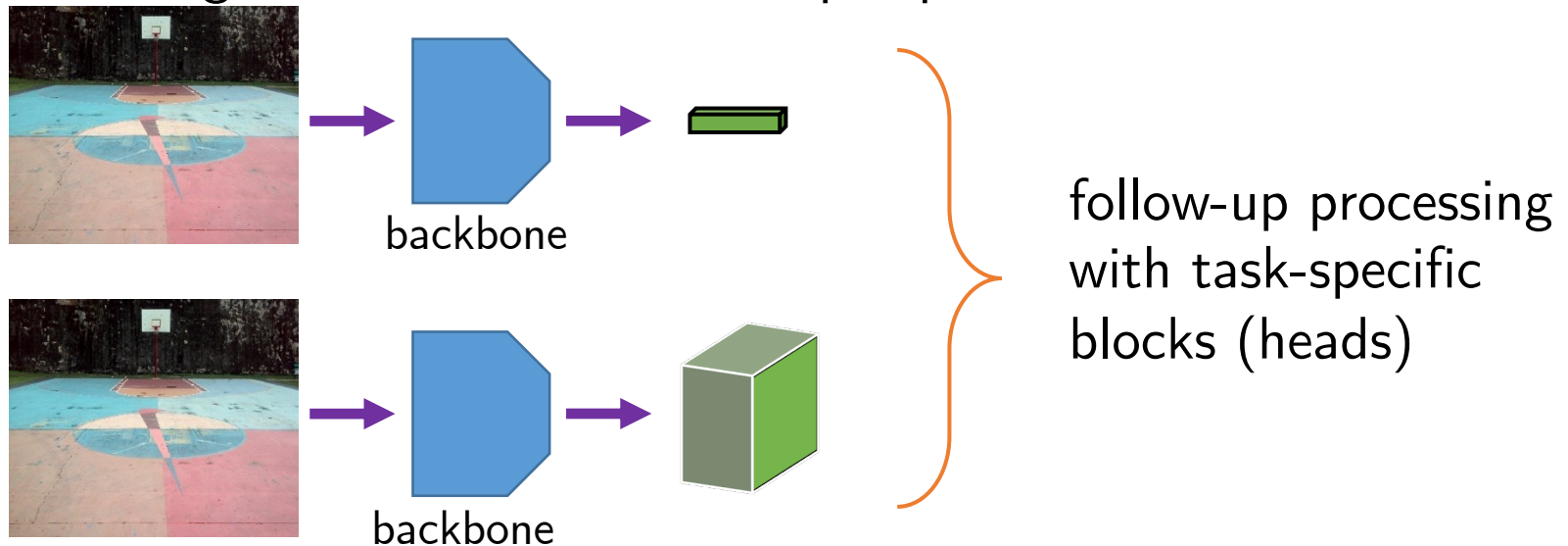
Computer Vision Methods 2024

CTU in Prague

Giorgos Tolias

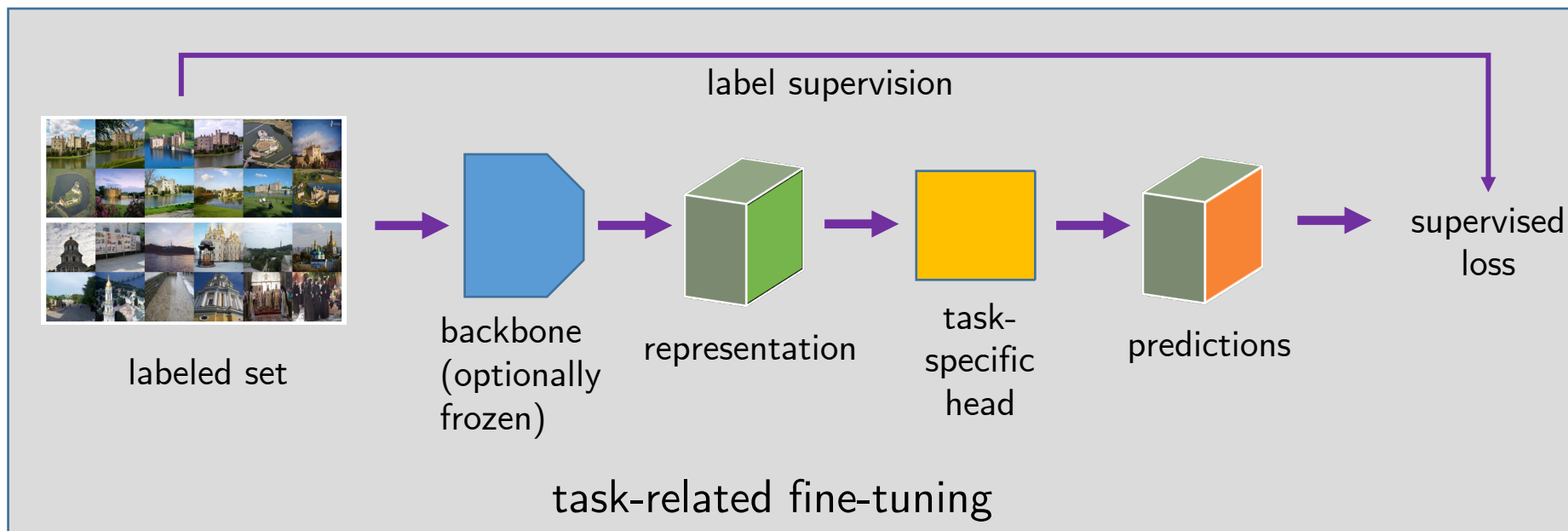
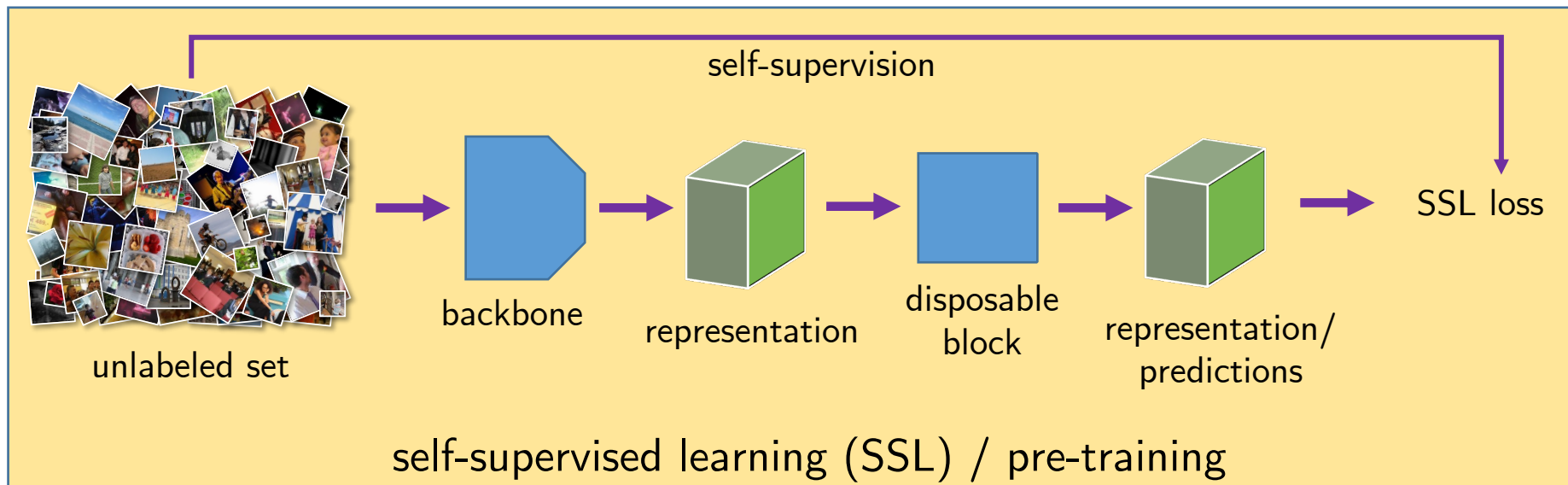
visual representation

- visual representation: at the core of every computer vision task
 - global image representation – single vector
 - retrieval, classification
 - dense image representation - 3D tensor
 - segmentation, detection, depth prediction



- the backbone requires lots of training data → use self-supervision
 - train one backbone and use for many final tasks
- heads can be trained with fewer data given a good representation

SSL pre-training and task-related fine-tuning

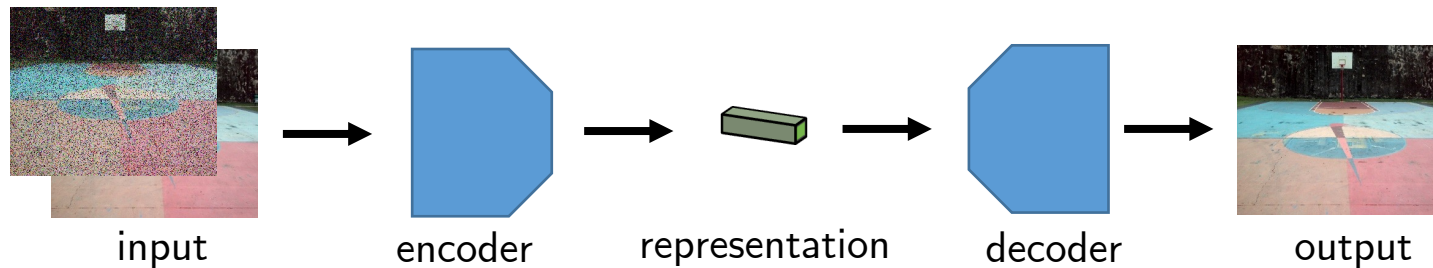


SSL method types

- auto-encoders
 - input is the output label
 - concealed-restored information
 - concealed part of the input is the output label
 - exploit spatial/temporal context
 - relation between parts of the input is the label
 - instance discrimination / multi-view invariance
 - each image constitutes a class
 - image collection processing
 - joint processing of many images provides labels
 - discovery of repeating patterns/parts/objects
- non-exhaustive listing of method types
 - non-exhaustive listing of methods – the literature is vast

auto-encoders

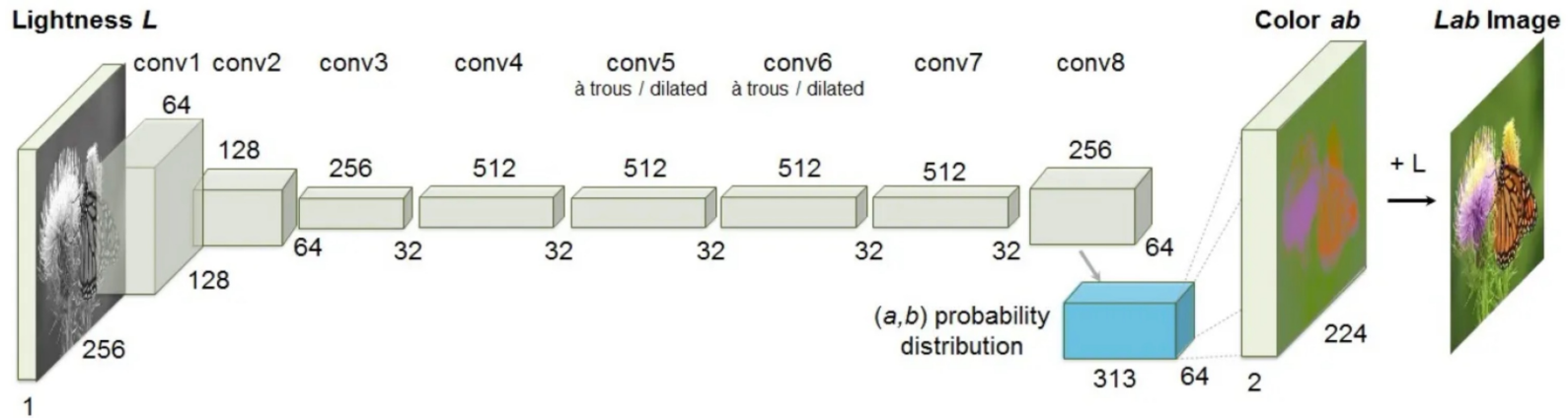
auto-encoder



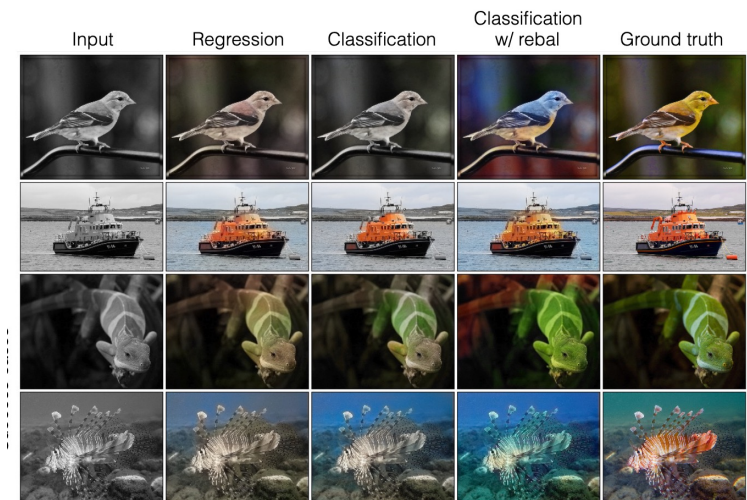
- reconstruct the input at the output
 - pixel-wise mean-squared error as loss
- reconstruction works → representation captures structures/regularities of the input data
- denoising auto-encoder
 - add noise to the input: remove the effect of stochastic corruption
 - better representation

concealed-restored
information

colorization

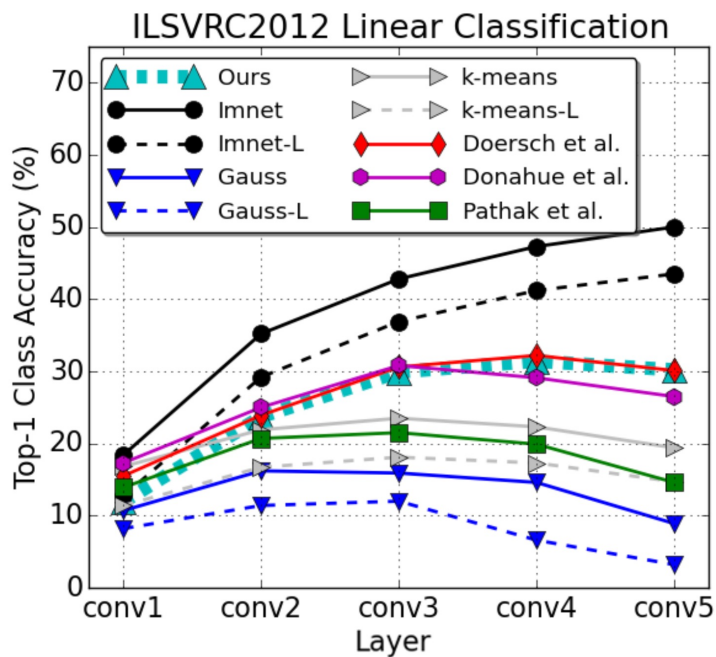


- use Lab color space
- L in the input $\rightarrow$ predict ab at the output
- cast as classification task
 - better than regression
- the better the colorization performance
 - the better the internal representation



colorization

- the last layer is not the best at generalization
- use the learned representation and fine-tune for other tasks
 - classification
 - detection
 - segmentation



Dataset and Task Generalization on PASCAL [37]								
fine-tune layers	[Ref]	Class. (%mAP)			Det. (%mAP)		Seg. (%mIU)	
		fc8	fc6-8	all	[Ref]	all	[Ref]	all
ImageNet [38]	-	76.8	78.9	79.9	[36]	56.8	[42]	48.0
Gaussian	[10]	-	-	53.3	[10]	43.4	[10]	19.8
Autoencoder	[16]	24.8	16.0	53.8	[10]	41.9	[10]	25.2
k-means [36]	[16]	32.0	39.2	56.6	[36]	45.6	[16]	32.6
Ours (gray)	-	52.4	61.5	65.9	-	46.1	-	35.0
Ours (color)	-	52.4	61.5	65.6	-	46.9	-	35.6

[Zhang, Colorful Image Colorization, ECCV2016]

inpainting

- hide a part of the input $\rightarrow$ reconstruct it at the output
- reconstruction loss + adversarial loss (sharper results)
 - discriminator network predicts whether an image is real or synthetic
 - two player game as in GANs



(a) Input context

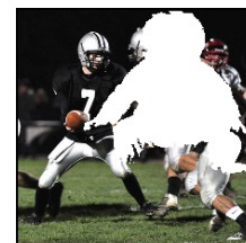
(b) Human artist

(c) Context Encoder
(L_2 loss)(d) Context Encoder
(L_2 + Adversarial loss)

(a) Central region



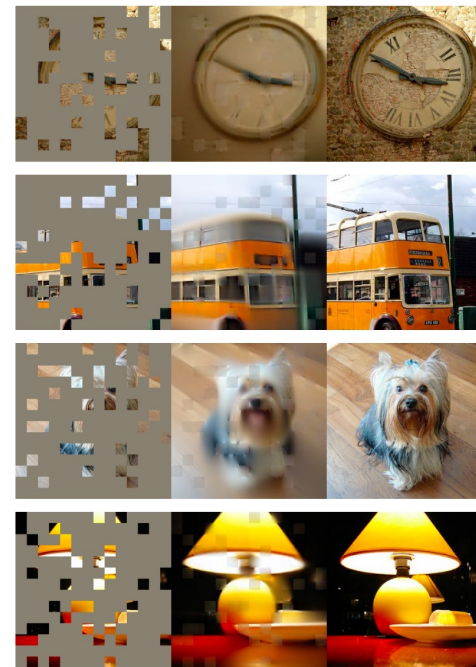
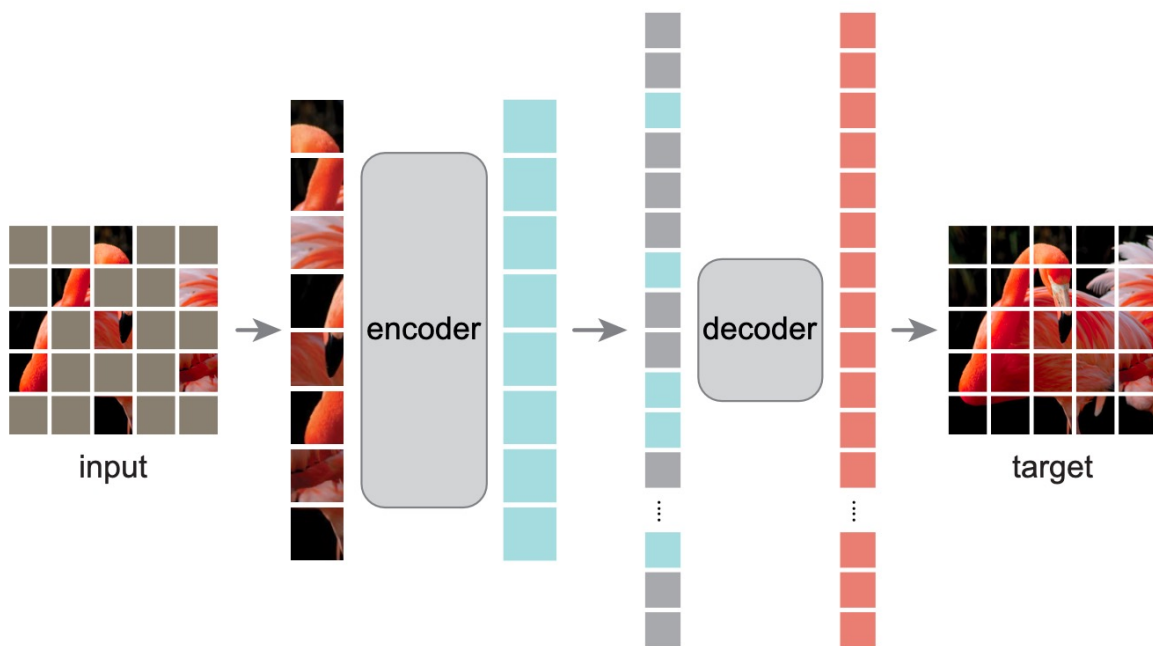
(b) Random block



(c) Random region

[Pathak, Context Encoders: Feature Learning by Inpainting, CVPR2016]

masked auto-encoder (mae)

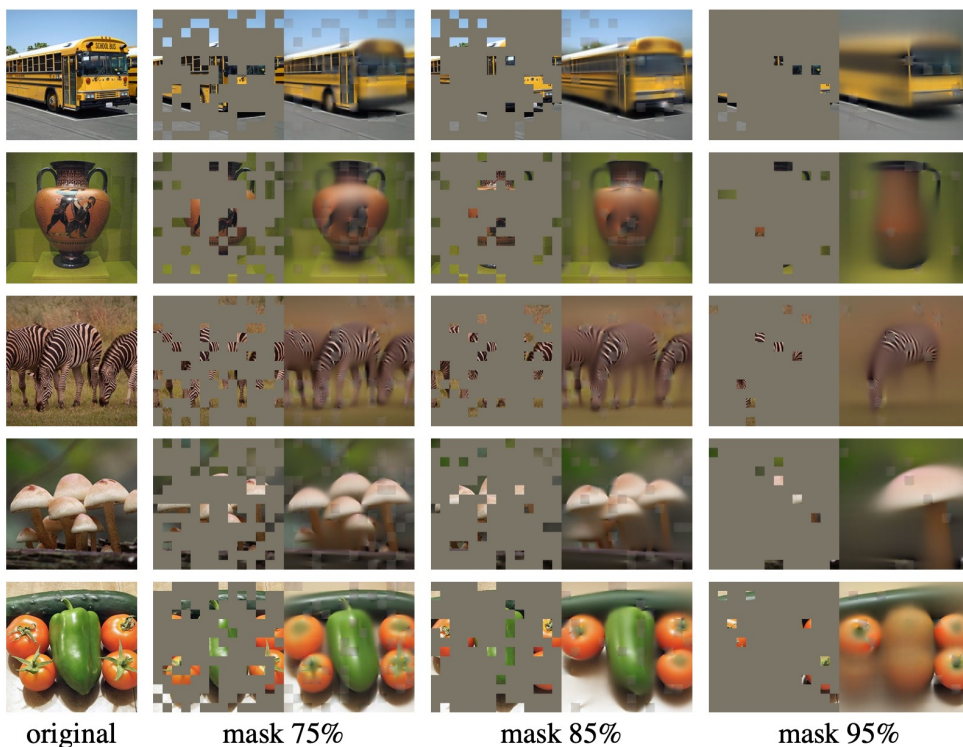


- similar to denoising auto-encoders / inpainting: tailored for ViT
 - mask-out some of the input patches
 - reconstruct them at the output
- asymmetric encoder-decoder operation
 - encoder: operates on non masked patches
 - decoder: operates on all patches – learnable mask token included

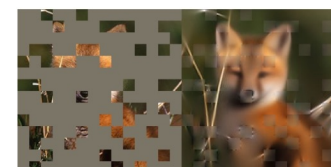
[He, Masked Autoencoders Are Scalable Vision Learners, CVPR2022]

masked auto-encoders (mae)

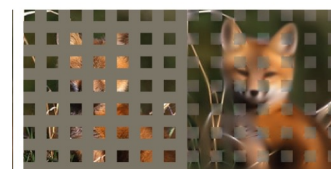
- do not allow to solve by extrapolation of visible patches
 - large masking ratio
 - random samples for masking
- even if wrong, semantically plausible reconstructions
 - representation captures semantics



block 50%

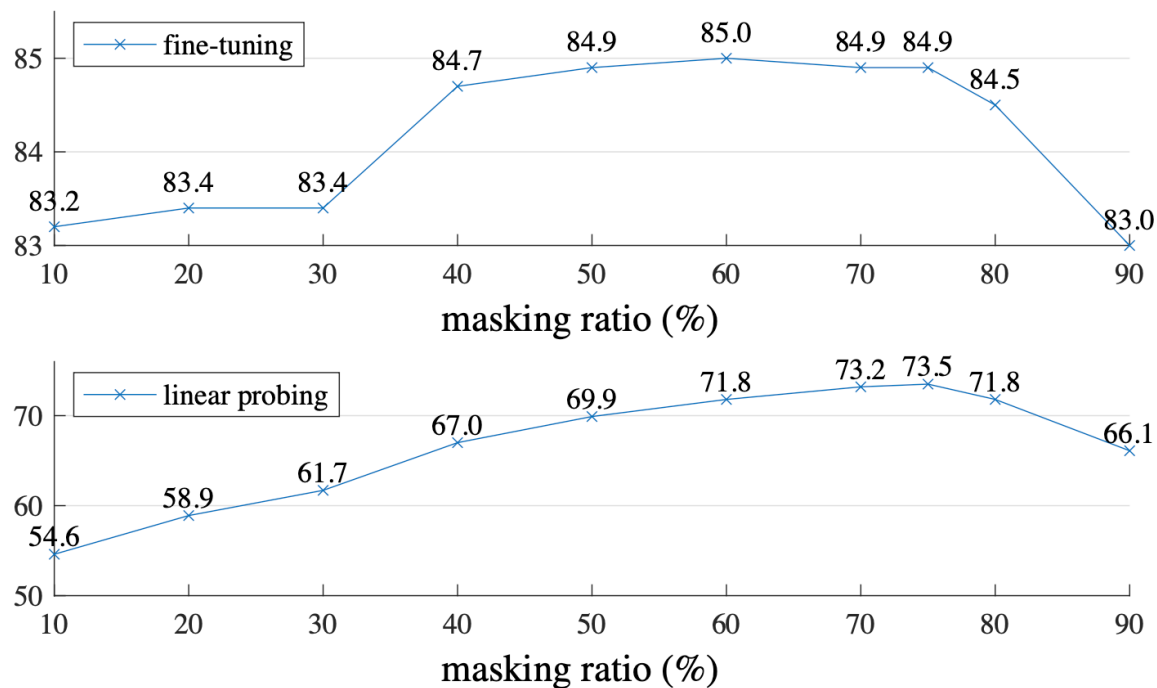


random 75%



grid 75%

masked auto-encoder (mae)



blocks	ft	lin
1	84.8	65.5
2	84.9	70.0
4	84.9	71.9
8	84.9	73.5
12	84.4	73.3

(a) **Decoder depth.** A deep decoder can improve linear probing accuracy.

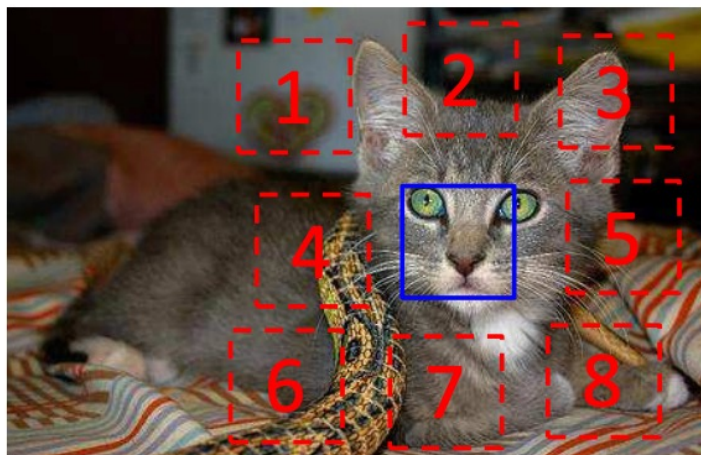
case	ratio	ft	lin
random	75	84.9	73.5
block	50	83.9	72.3
block	75	82.8	63.9
grid	75	84.0	66.0

(f) **Mask sampling.** Random sampling works the best. See Figure 6 for visualizations.

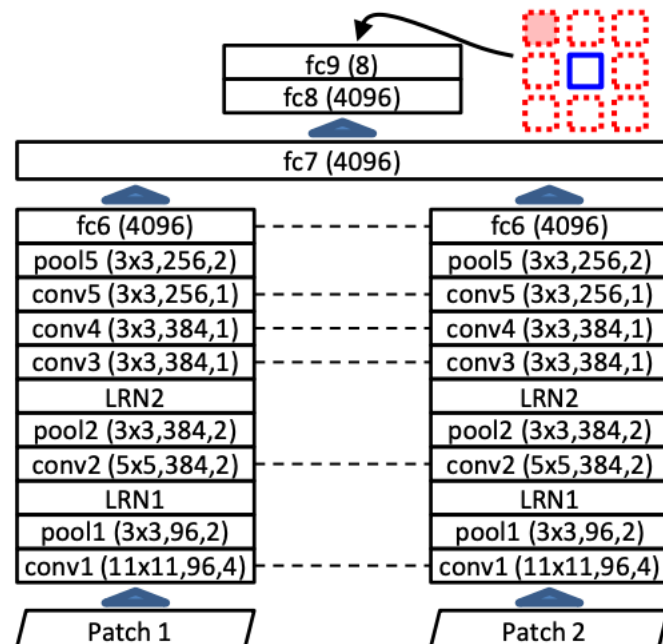
exploit spatial/temporal
context

context prediction

- choose a deformed 3 × 3 grid of image regions
- input: central region + a second region
→ predict the position of the second region

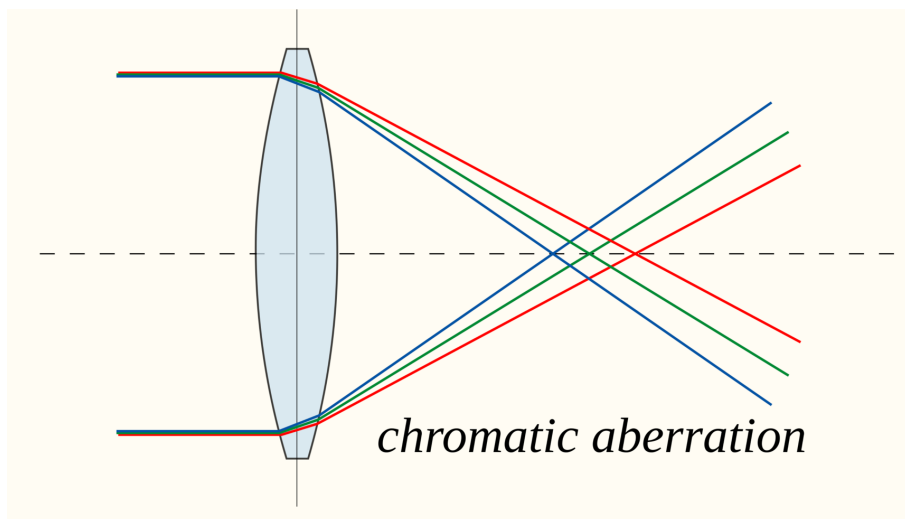


$$X = \left(\begin{array}{c} \text{central region} \\ \text{second region} \end{array} \right); Y = 3$$



context prediction

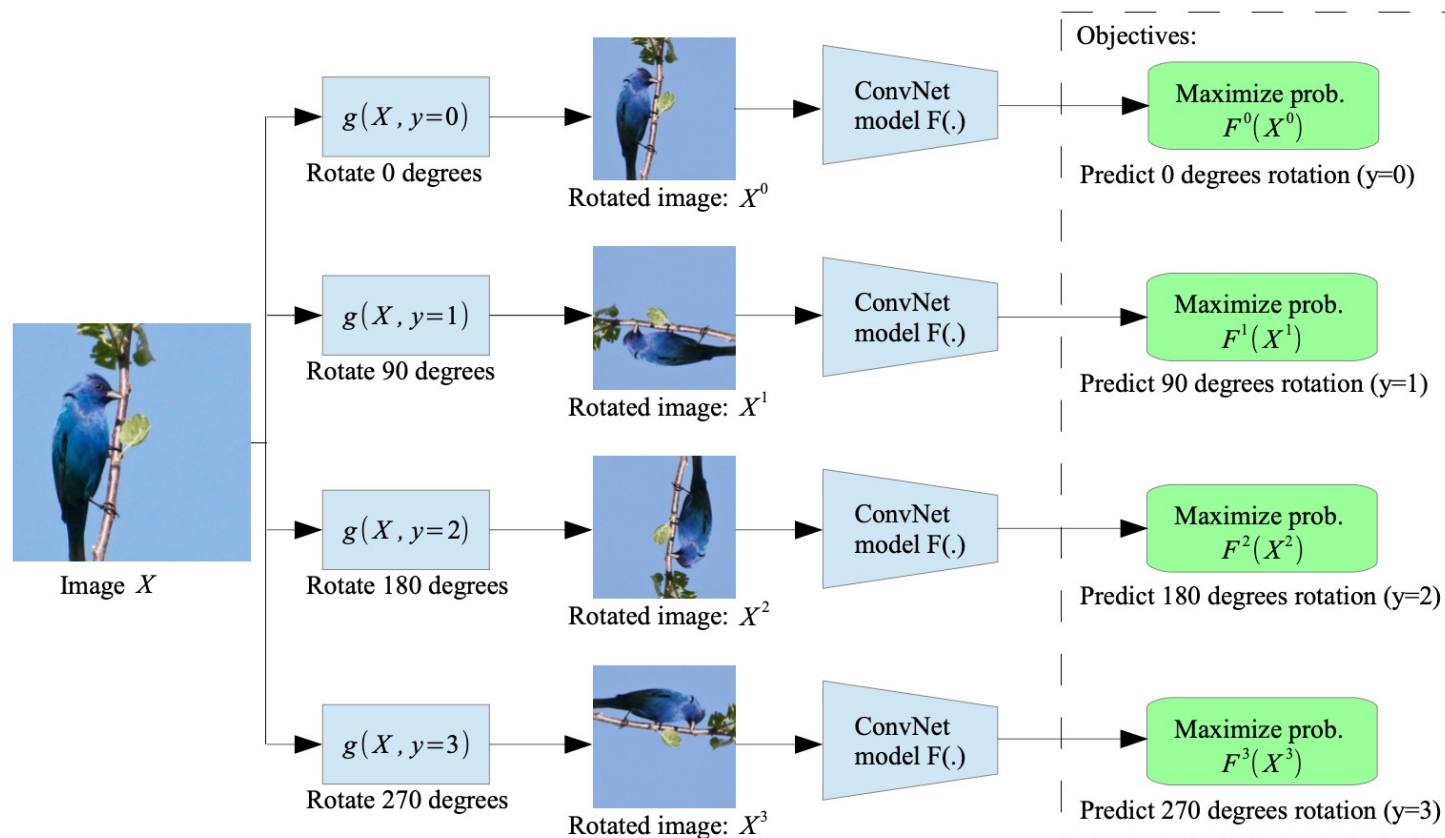
- **chromatic aberration:** failure of a lens to focus all colors to the same point



- deep networks are good at finding solution shortcuts
 - task solved through learning chromatic aberration patterns
 - no useful representation learned
 - skip 2 out of 3 color channels
 - good representation learned

predicting object rotations (rotnet)

- assumption: people typically take photos of upright objects
- rotate images and use object orientation as label



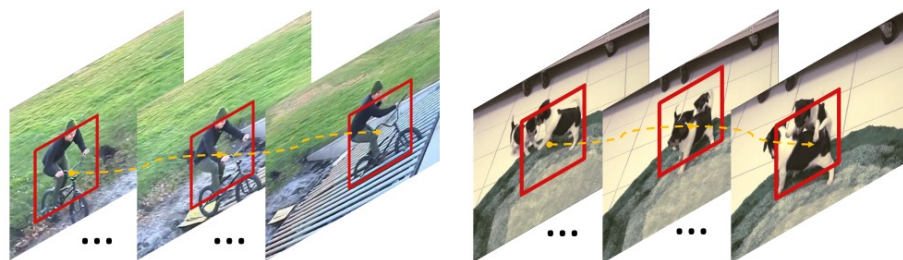
predicting object rotations (rotnet)

# Rotations	Rotations	CIFAR-10 Classification Accuracy
4	0°, 90°, 180°, 270°	89.06
8	0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°	88.51
2	0°, 180°	87.46
2	90°, 270°	85.52

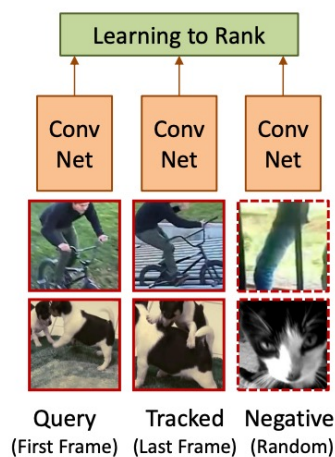
Method	Conv4	Conv5
ImageNet labels from (Bojanowski & Joulin, 2017)	59.7	59.7
Random from (Noroozi & Favaro, 2016)	27.1	12.0
Tracking Wang & Gupta (2015)	38.8	29.8
Context (Doersch et al., 2015)	45.6	30.4
Colorization (Zhang et al., 2016a)	40.7	35.2
Jigsaw Puzzles (Noroozi & Favaro, 2016)	45.3	34.6
BIGAN (Donahue et al., 2016)	41.9	32.2
NAT (Bojanowski & Joulin, 2017)	-	36.0
(Ours) RotNet	50.0	43.8

video tracking for positive examples

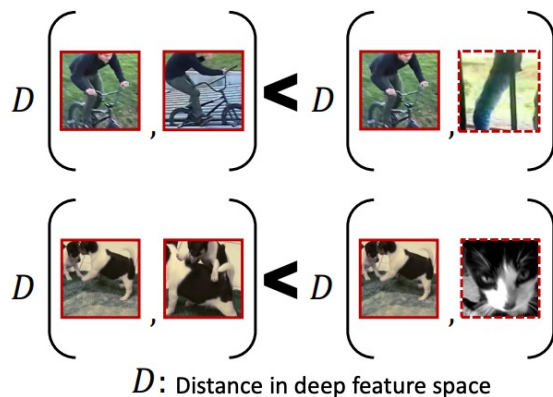
- representation learning with triplet loss: as in deep metric learning
- use video frames as training data
 - negatives: frame region far enough in time
 - positives: output of a video tracking algorithm



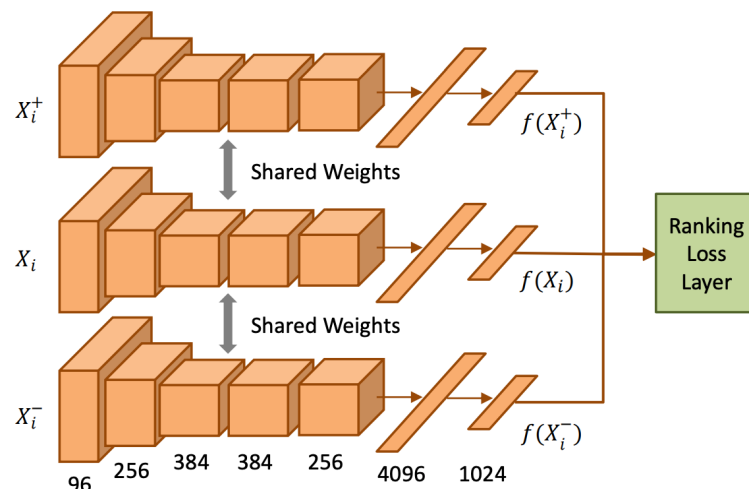
(a) Unsupervised Tracking in Videos



(b) Siamese-triplet Network



(c) Ranking Objective

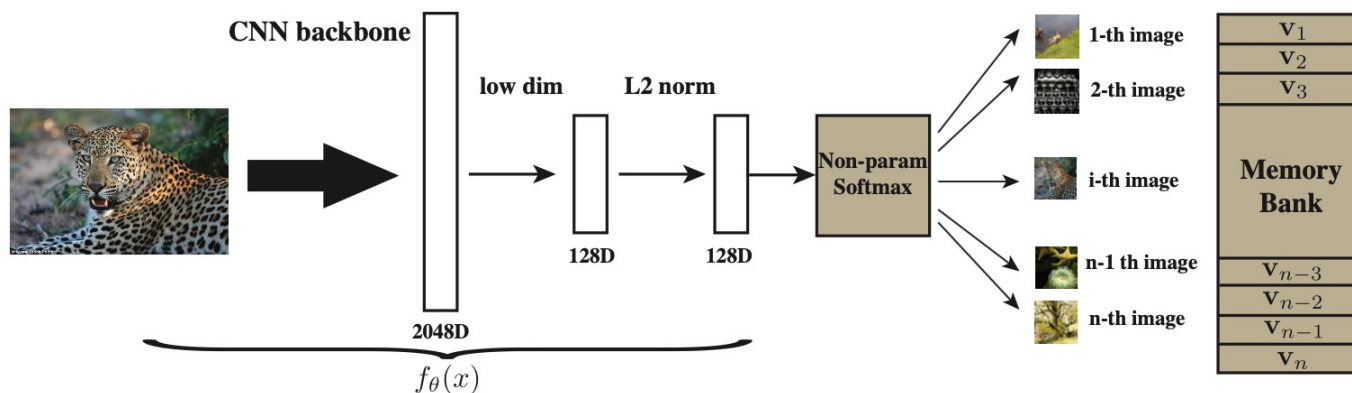


instance discrimination
multi-view invariance

instance discrimination

- each image (and all of its augmentations) forms a class by itself
- optimize non-parametric soft-max classifier
 - a type of contrastive loss
- cannot process all training images in a batch
 - memory of previously extracted representations
 - works despite inconsistency of the representation extraction

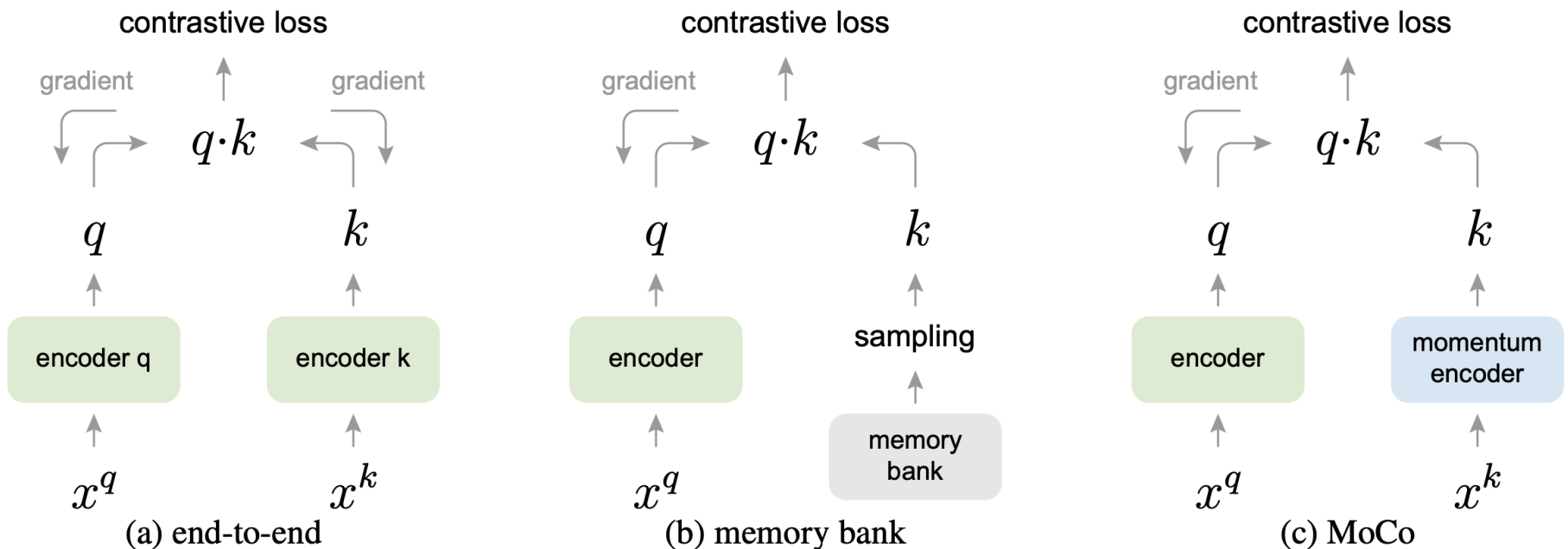
$$P(i|\mathbf{v}) = \frac{\exp(\mathbf{v}_i^T \mathbf{v} / \tau)}{\sum_{j=1}^n \exp(\mathbf{v}_j^T \mathbf{v} / \tau)}$$



Training / Testing	Linear SVM	Nearest Neighbor
Param Softmax	60.3	63.0
Non-Param Softmax	75.4	80.8
NCE $m = 1$	44.3	42.5
NCE $m = 10$	60.2	63.4
NCE $m = 512$	64.3	78.4
NCE $m = 4096$	70.2	80.4

momentum contrast - moco

- train with contrastive loss for instance discrimination
 - end-to-end: memory size coupled with mini-batch size
 - memory bank: old entry inconsistency & no memory back-prop
 - queue & momentum encoder: $\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$
 - reuse examples from the preceding mini-batches
 - slowly progressing encoder for the memory



Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

```

# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
    x_q = aug(x) # a randomly augmented version
    x_k = aug(x) # another randomly augmented version

    q = f_q.forward(x_q) # queries: NxC
    k = f_k.forward(x_k) # keys: NxC
    k = k.detach() # no gradient to keys

    # positive logits: Nx1
    l_pos = bmm(q.view(N,1,C), k.view(N,C,1))

    # negative logits: NxK
    l_neg = mm(q.view(N,C), queue.view(C,K))

    # logits: Nx(1+K)
    logits = cat([l_pos, l_neg], dim=1)

    # contrastive loss, Eqn.(1)
    labels = zeros(N) # positives are the 0-th
    loss = CrossEntropyLoss(logits/t, labels)

    # SGD update: query network
    loss.backward()
    update(f_q.params)

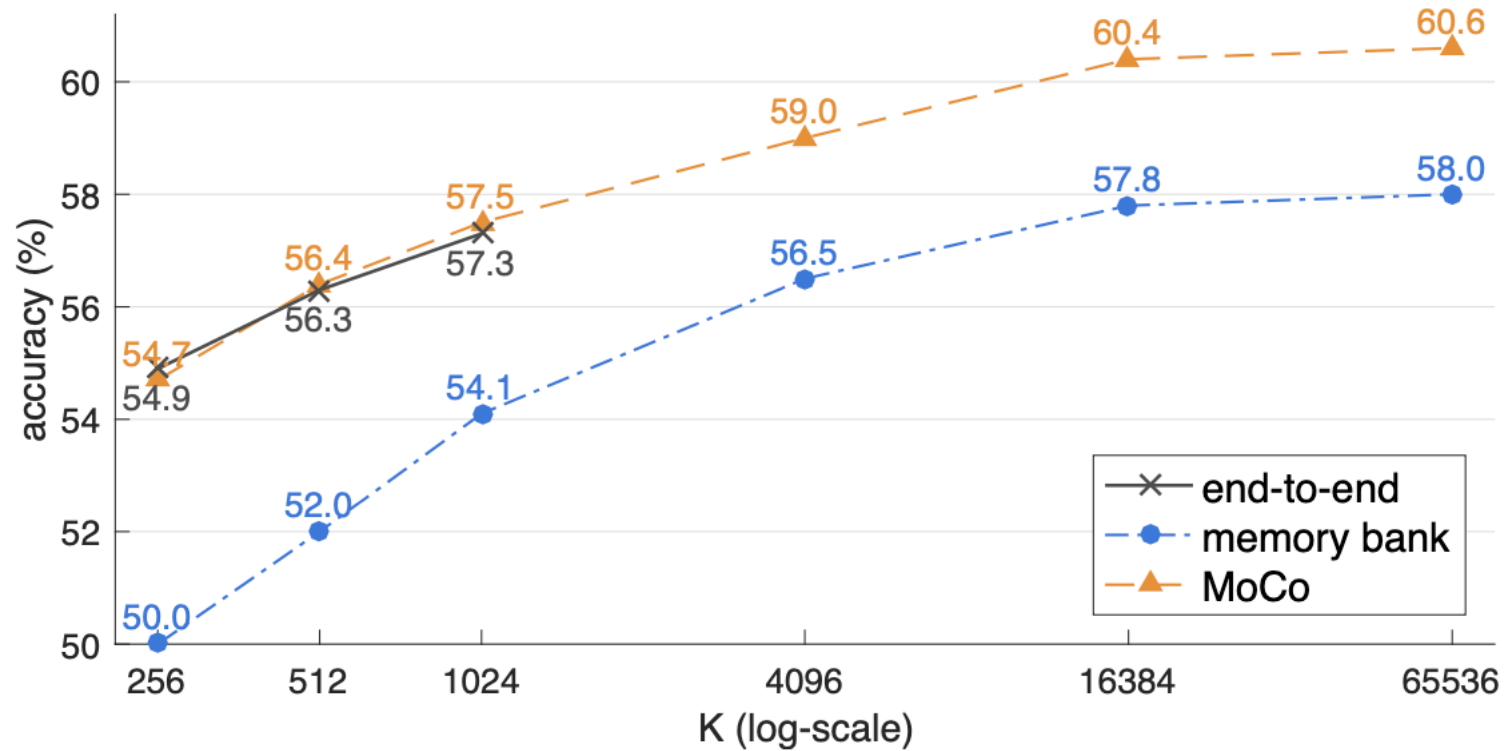
    # momentum update: key network
    f_k.params = m*f_k.params+(1-m)*f_q.params

    # update dictionary
    enqueue(queue, k) # enqueue the current minibatch
    dequeue(queue) # dequeue the earliest minibatch

```

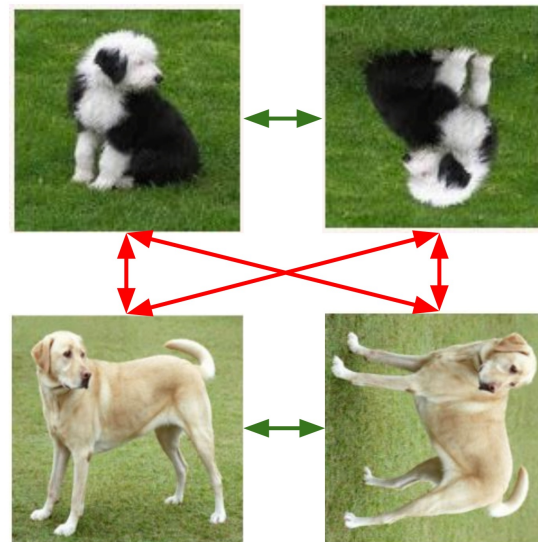
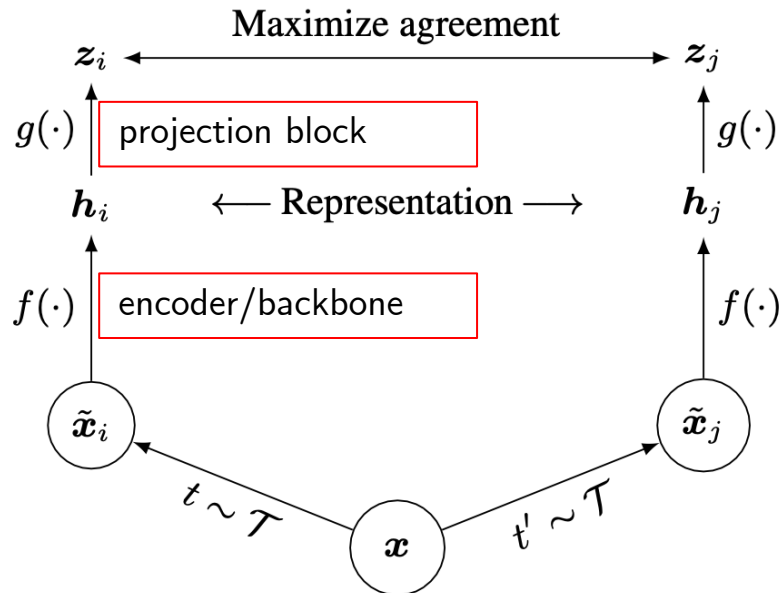
bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.

momentum contrast - moco

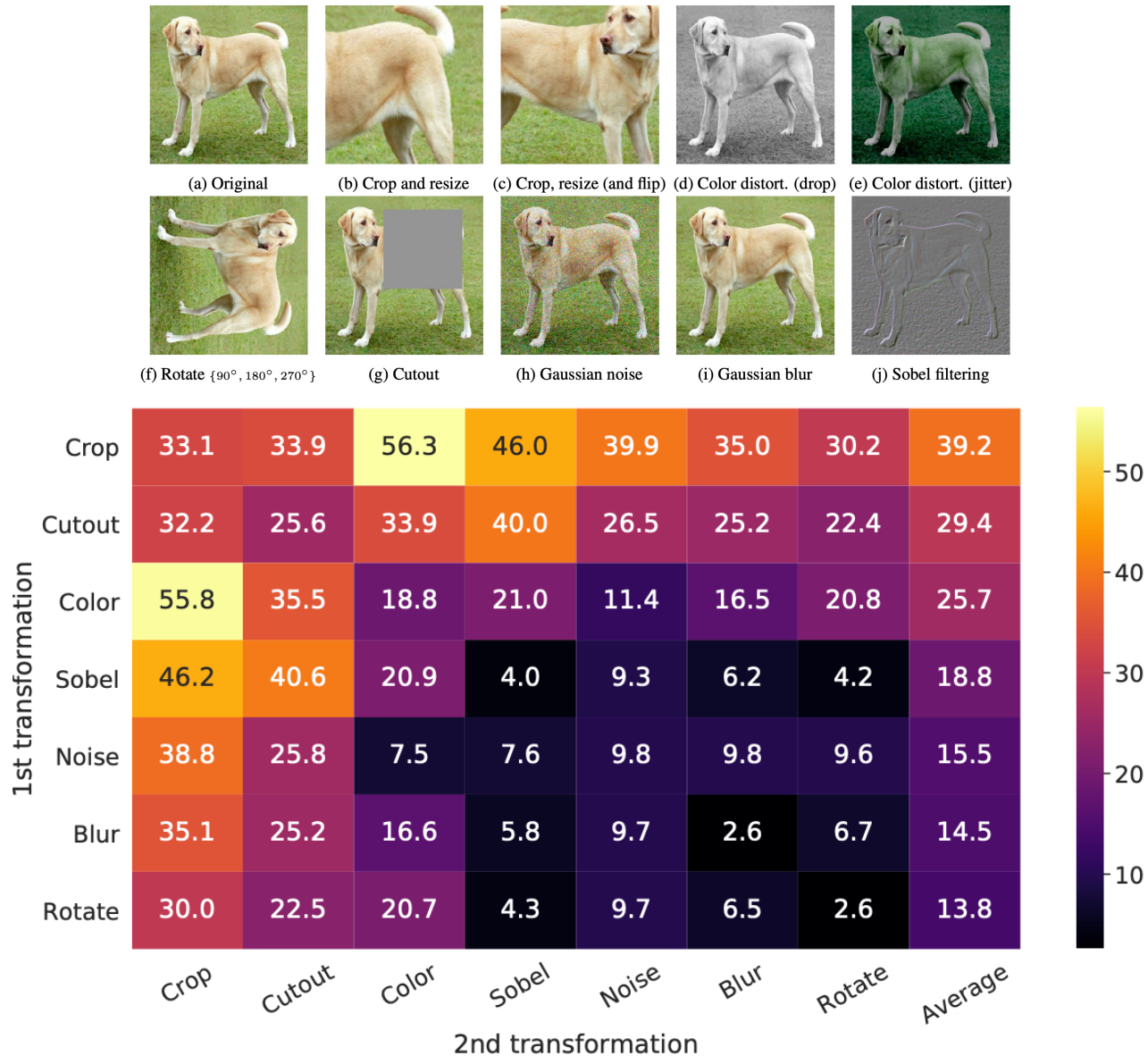


simple framework for contrastive learning - simclr

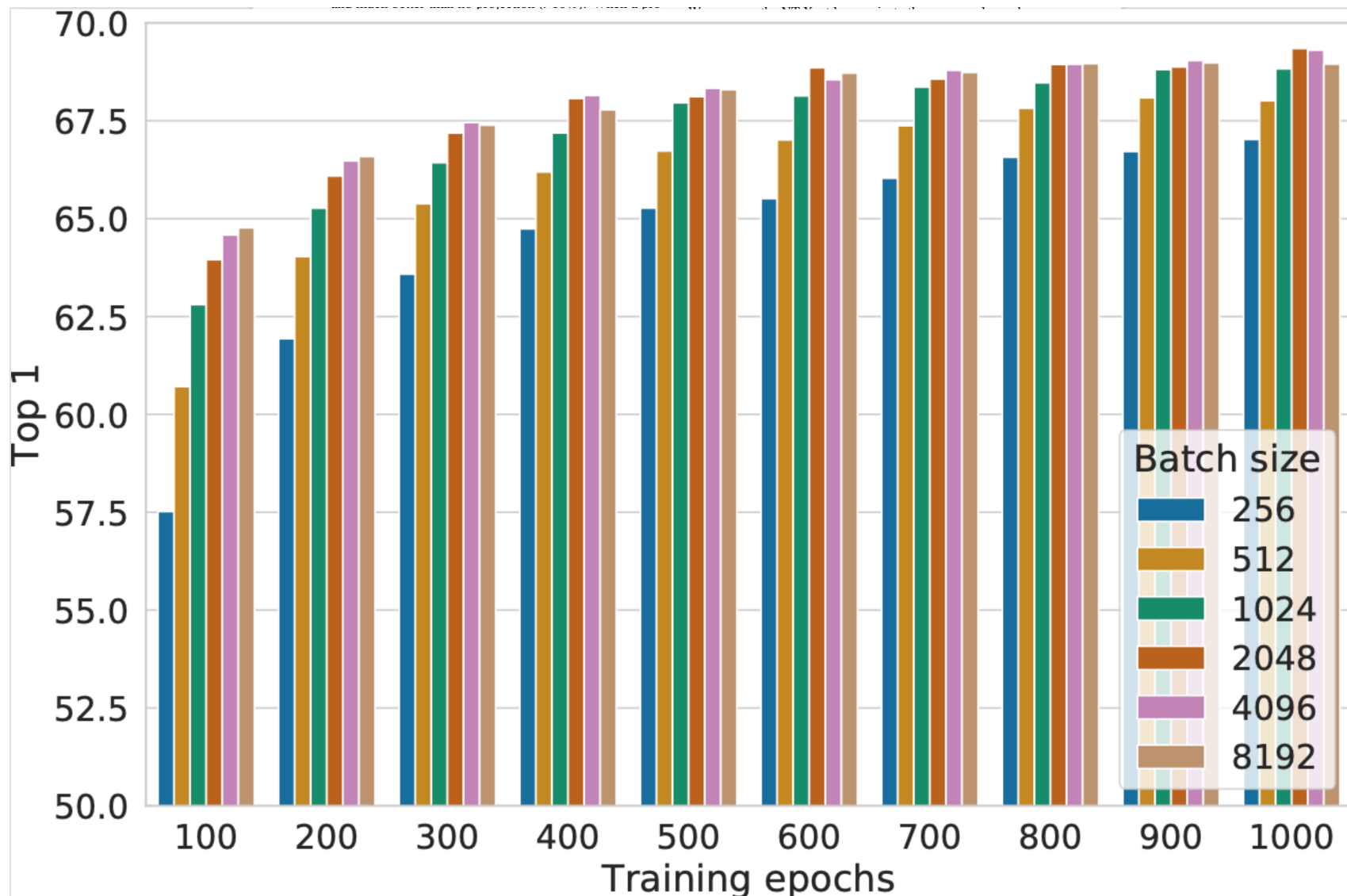
- end-to-end approach
 - no memory or asymmetric encoding
 - large batch through hardware support
- introduce disposable projection block
 - used for training and then removed
 - internal representation generalizes better
- composition of augmentations is a key



simple framework for contrastive learning - simclr



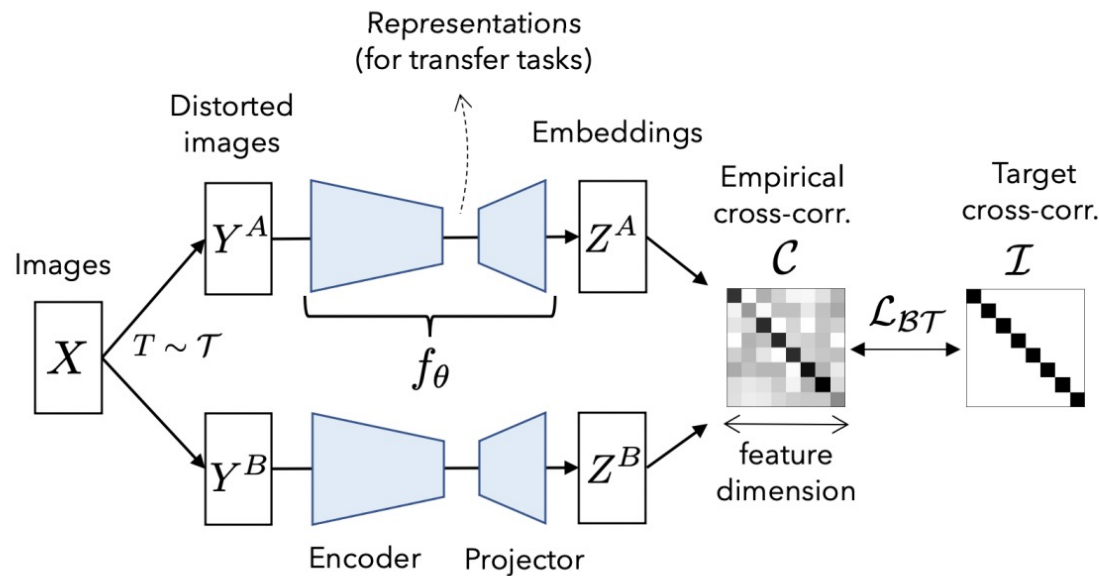
simple framework for contrastive learning - simclr



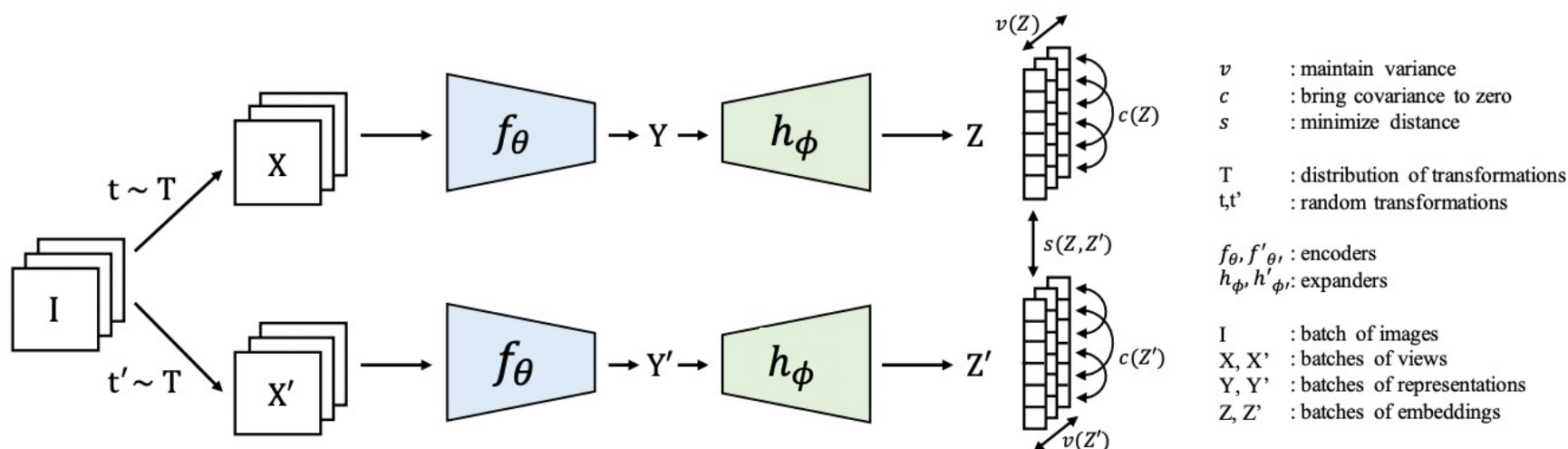
redundancy reduction - Barlow twins

- invariance term: standard term of matching the positives
- redundancy term: decorrelate the representation dimensions

$$\mathcal{L}_{\mathcal{BT}} \triangleq \underbrace{\sum_i (1 - C_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} C_{ij}^2}_{\text{redundancy reduction term}}$$



variance-invariance-covariance regularization - VICreg



- **Invariance:** the mean square distance between the embedding vectors.
- **Variance:** a hinge loss to maintain the standard deviation (over a batch) of each variable of the embedding above a given threshold. This term forces the embedding vectors of samples within a batch to be different.
- **Covariance:** a term that attracts the covariances (over a batch) between every pair of (centered) embedding variables towards zero. This term decorrelates the variables of each embedding and prevents an *informational collapse* in which the variables would vary together or be highly correlated.

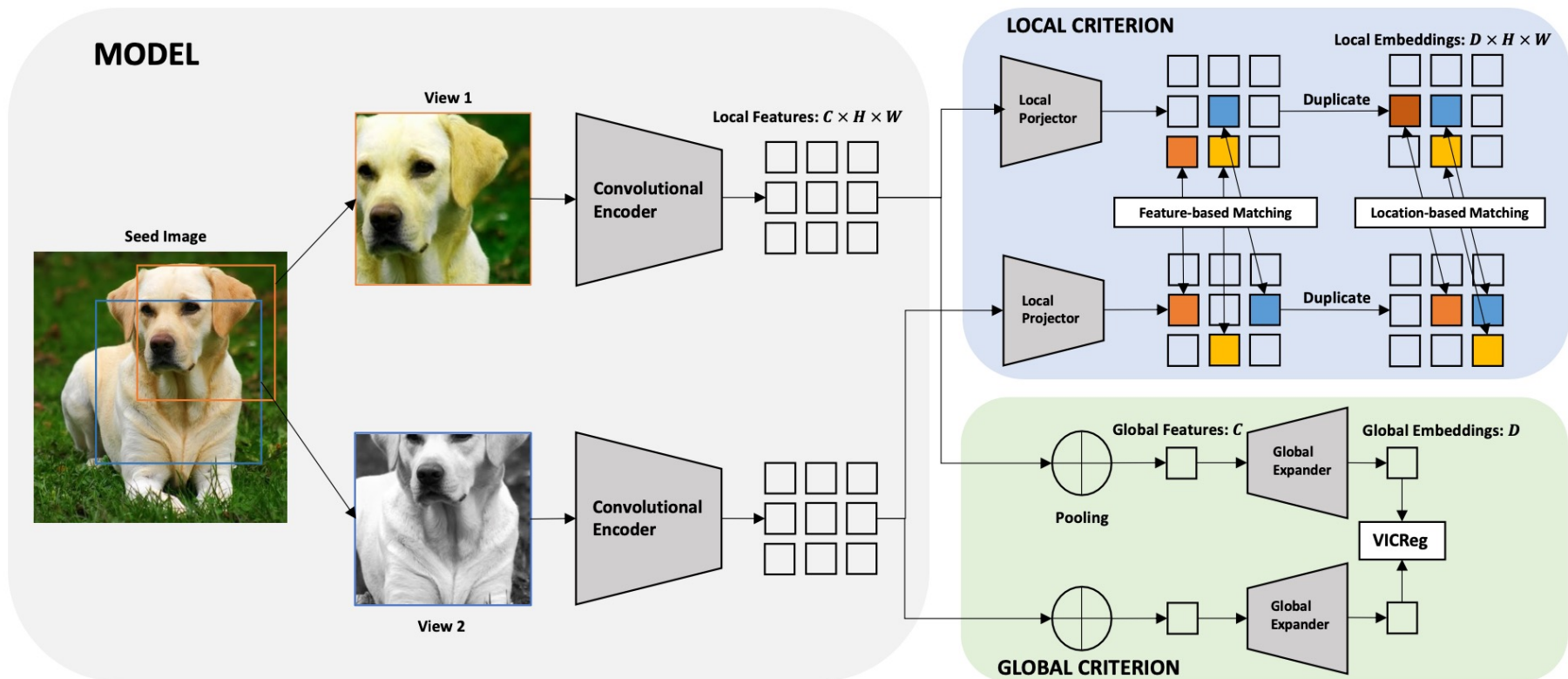
variance-invariance-covariance regularization - VICreg

- on par with previous approaches, but
 - simpler to implement
 - more intuitive

Method	Linear Classification			Object Detection		
	Places205	VOC07	iNat18	VOC07+12	COCO det	COCO seg
Supervised	53.2	87.5	46.7	81.3	39.0	35.4
MoCo [22]	46.9	79.8	31.5	-	-	-
PIRL [33]	49.8	81.1	34.1	-	-	-
SimCLR [8]	52.5	85.5	37.2	-	-	-
MoCo v2 [10]	51.8	86.4	38.6	82.5	39.8	36.1
SimSiam [11]	-	-	-	82.4	-	-
BYOL [21]	54.0	86.6	47.6	-	40.4 [†]	37.0 [†]
SwAV (w/ multi-crop) [7]	56.7	88.9	48.6	82.6	41.6	37.8
Barlow Twins [21]	54.1	86.2	46.5	82.6	40.0 [†]	36.7 [†]
VICReg (ours)	54.3	86.6	47.0	82.4	39.4	36.4

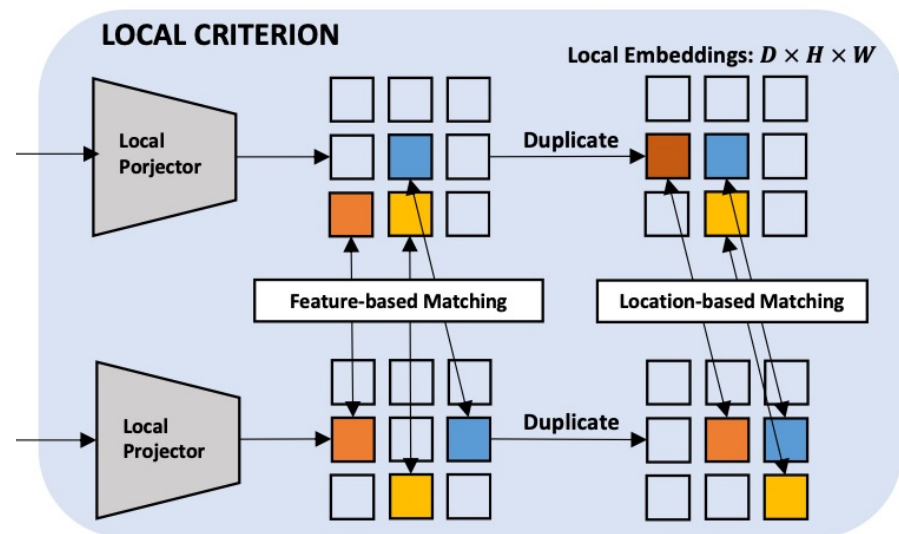
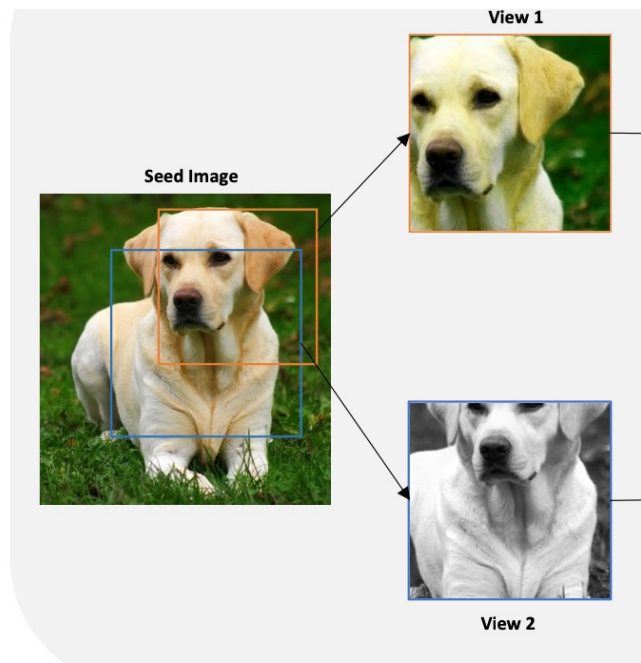
SSL of local representation - VICregL

- targets dense prediction tasks:
 - VICreg + loss on local representation



SSL of local representation - VICregL

- targets dense prediction tasks:
 - VICreg + loss on local representation
- corresponding/positive locations given by
 - the relative geometric transformation
 - neighbors in the representation space



SSL of local representation - VICRegL

- good performance for global and for dense prediction tasks

Method	Epochs	Linear Cls. (%)	Linear Seg. (mIoU)		
		ImageNet Frozen	Pascal VOC Frozen	Fine-Tuned	Cityscapes Frozen
<i>Global features</i>					
MoCo v2 [Chen et al., 2020b]	200	67.5	35.6	64.8	14.3
SimCLR [Chen et al., 2020a]	400	68.2	45.9	65.4	17.9
BYOL [Grill et al., 2020]	300	72.3	47.1	65.7	22.6
VICReg [Bardes et al., 2022]	300	71.5	47.8	65.5	23.5
<i>Local features</i>					
PixPro [Xie et al., 2021]	400	60.6	52.8	67.5	22.6
DenseCL [Wang et al., 2021]	200	65.0	45.3	66.8	11.2
DetCon [Hénaff et al., 2021]	1000	66.3	53.6	67.4	16.2
InsLoc [Yang et al., 2022]	400	45.0	24.1	64.4	7.0
CP ² [Wang et al., 2022]	820	53.1	21.7	65.2	8.4
ReSim [Xiao et al., 2021]	400	59.5	51.9	67.3	12.3
<i>Ours</i>					
VICRegL $\alpha = 0.9$	300	71.2	54.0	66.6	25.1
VICRegL $\alpha = 0.75$	300	70.4	55.9	67.6	25.2

[Bardes, VICRegL: Self-Supervised Learning of Local Visual Features, NeurIPS2022]

object discovery and dense representation networks

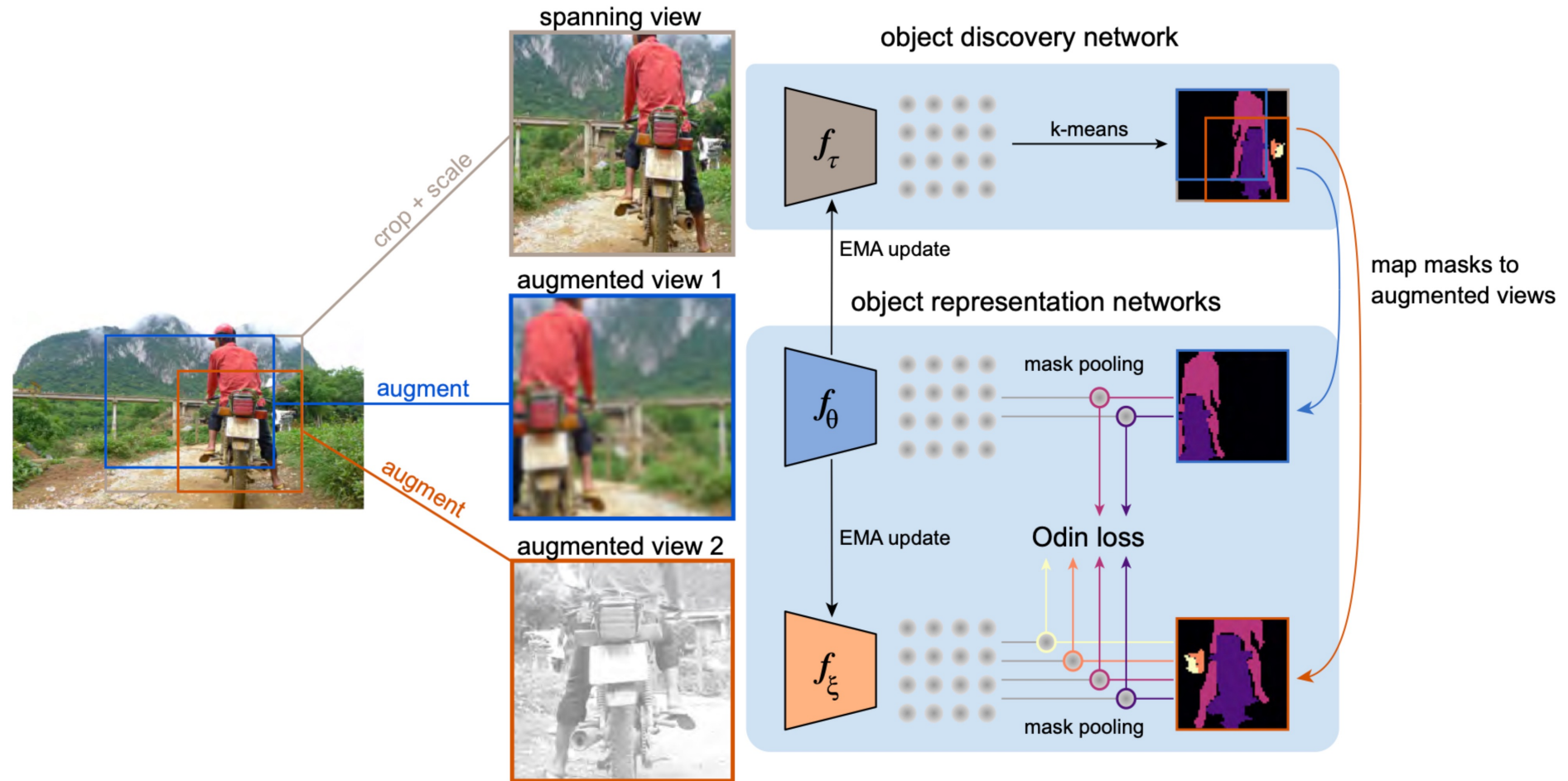
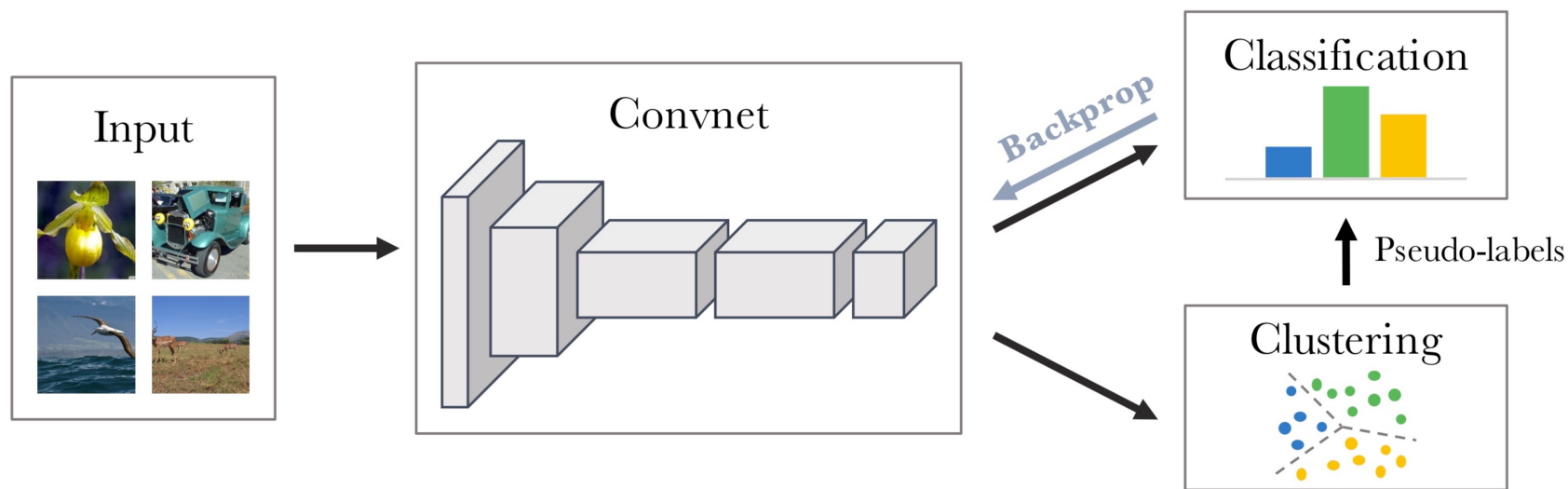


image collection processing

deep clusters as supervision

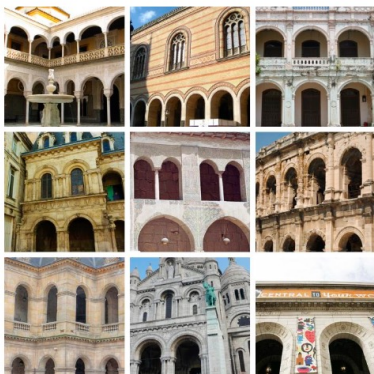
- assumption: common objects repeat in the training set
- clustering of the current representation $\rightarrow$ pseudo-labels
- iterative process (network and pseudo-labels get improved):
 - clustering $\rightarrow$ training $\rightarrow$ clustering $\rightarrow$...
- no domain knowledge
 - does not rely on well designed augmentations for positives



[Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze , Deep Clustering for Unsupervised Learning of Visual Features, ECCV2018]

deep clusters as supervision

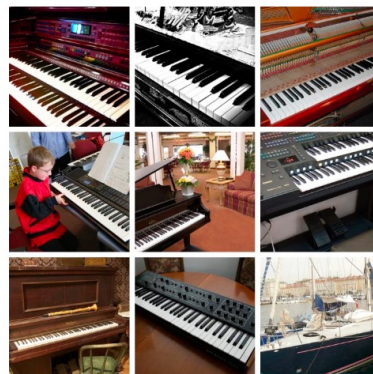
Filter 0



Filter 33



Filter 145



Filter 194



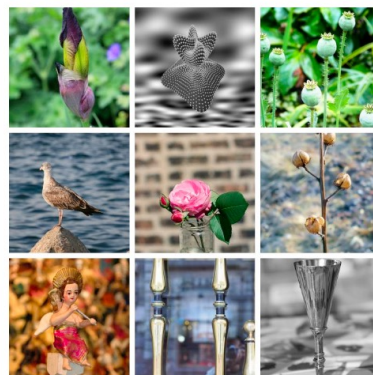
Filter 97



Filter 116



Filter 119



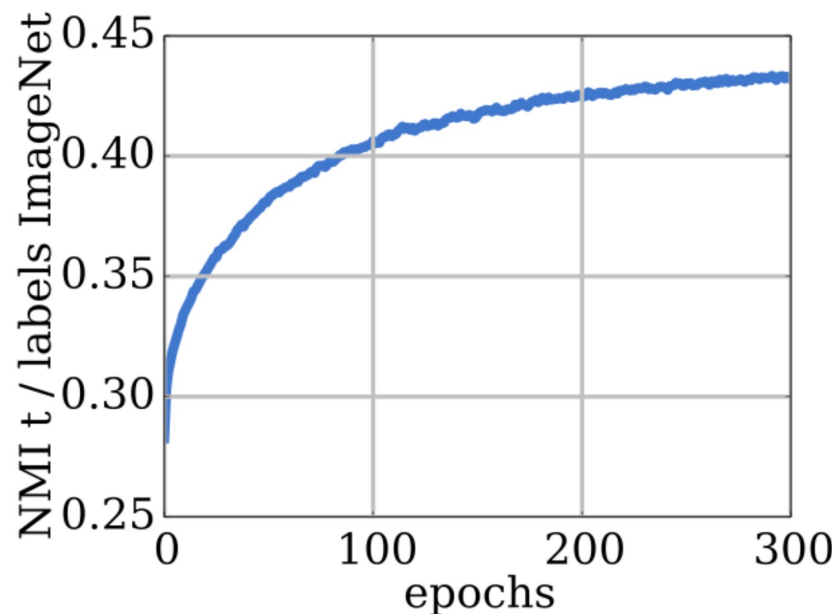
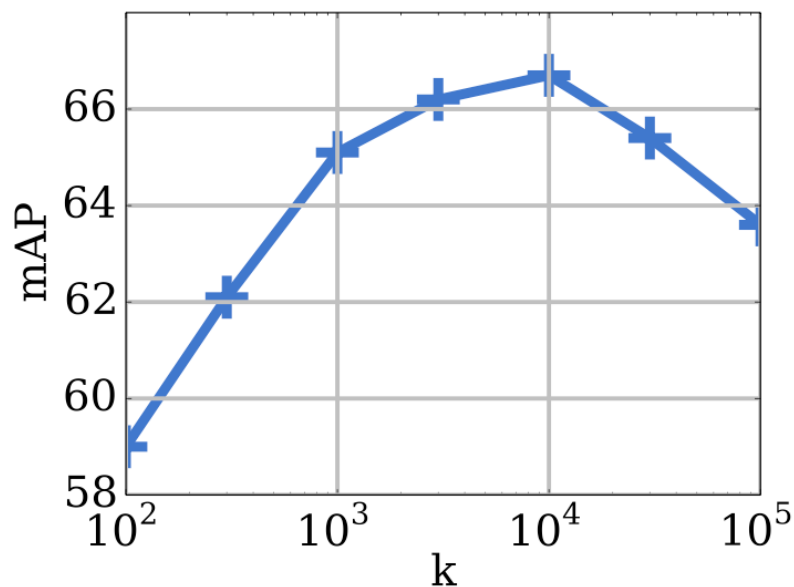
Filter 182



[Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze , Deep Clustering for Unsupervised Learning of Visual Features, ECCV2018]

deep clusters as supervision

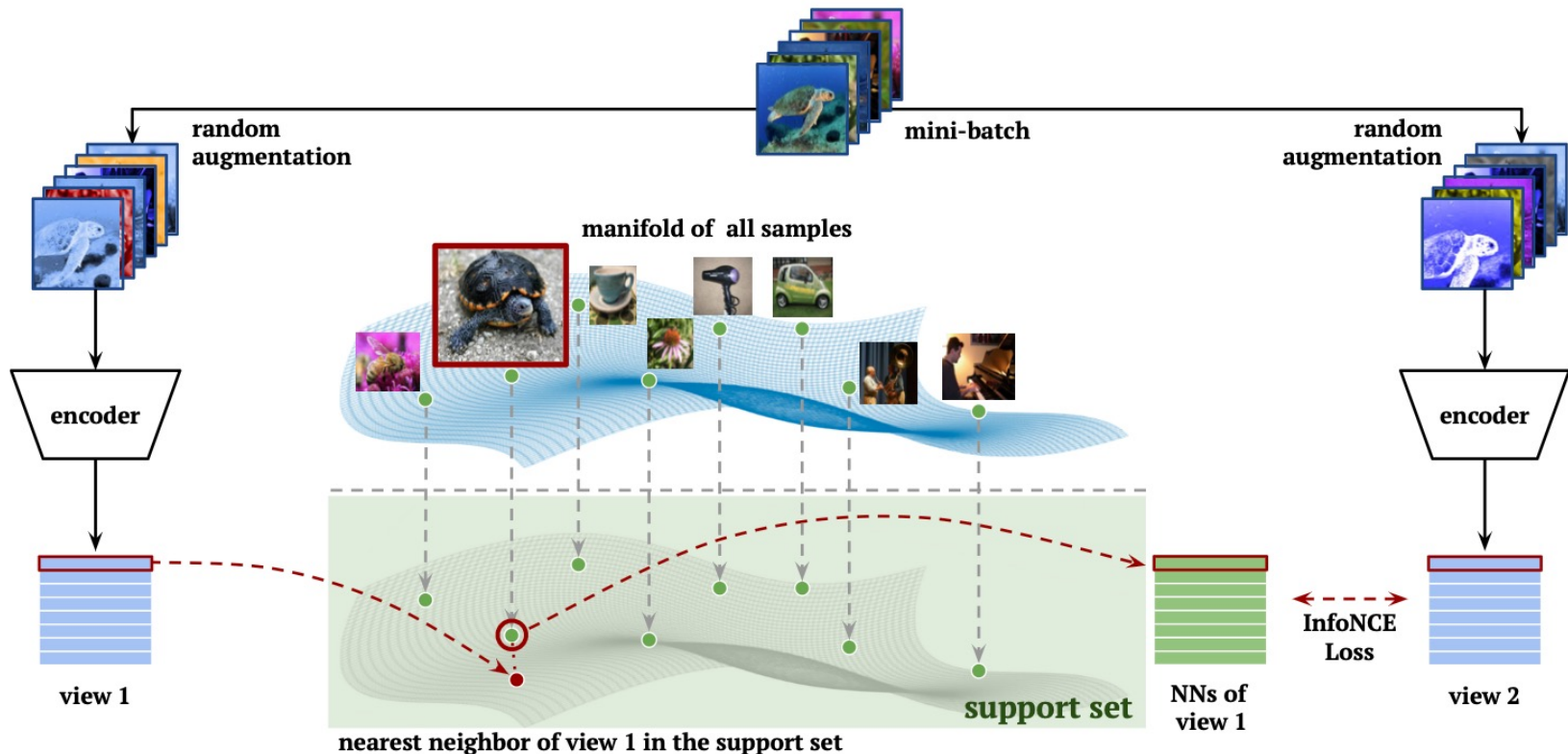
- intermediate number of clusters is the optimal
- mutual information between clusters and class labels increases



[Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze , Deep Clustering for Unsupervised Learning of Visual Features, ECCV2018]

neighbors as positives

- assumption: repeating objects
 - use them for positives: realistic compared to augmentations



[Dwibedi, With a Little Help from My Friends: Nearest-Neighbor Contrastive Learning of Visual Representations]

neighbors as positives

- search requires same representation model for query and database
 - re-processing takes time
 - using previously processed examples: inconsistent representation
- use queue with momentum encoder

Queue Size	8192	16384	32768	65536	98304
Top-1	73.6	74.2	74.9	75.0	75.4
Top-5	91.2	91.7	92.1	92.2	92.3

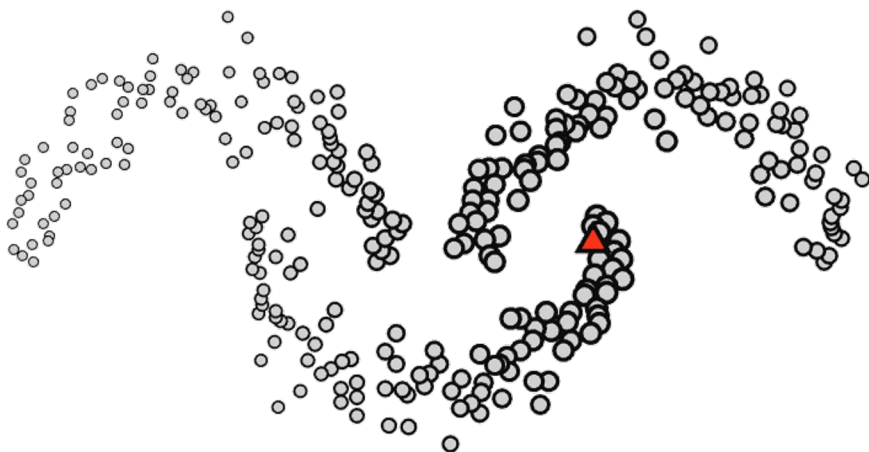
k in k -NN	1	2	4	8	16	32
Top-1	74.9	74.1	73.8	73.8	73.8	73.2
Top-5	92.1	91.6	91.5	91.4	91.3	91.2

[Dwibedi, With a Little Help from My Friends: Nearest-Neighbor Contrastive Learning of Visual Representations, ICCV'21]

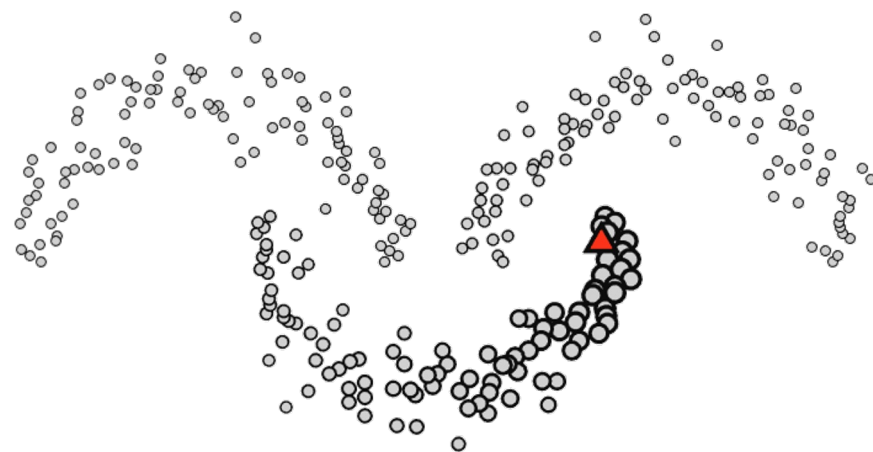
ssl for fine-tuning

mining (neighbors) on manifolds

- rely on manifold distance instead of Euclidean distance
- mine positives and negatives
- focus on hard-examples
 - differences between Euclidean and manifold neighbors



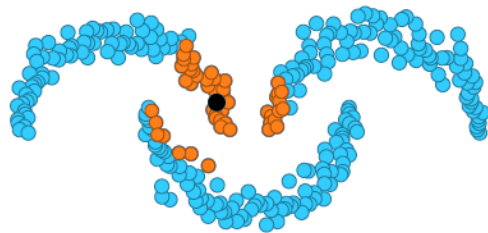
Euclidean



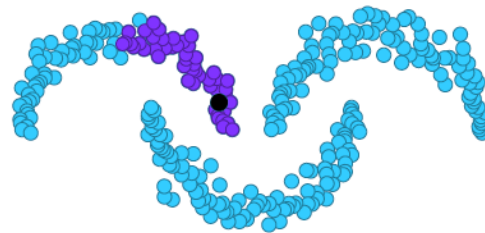
manifold

point size reflects similarity to query

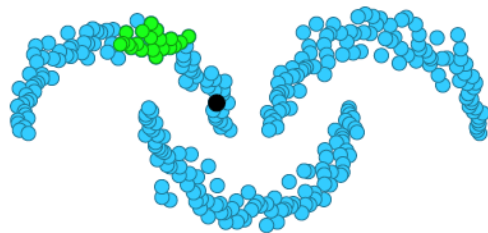
mining on manifolds



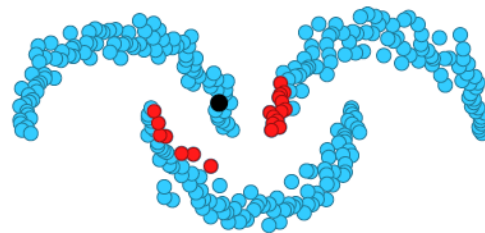
Euclidean NN (orange): $E(\mathbf{y})$



Manifold NN (purple): $M(\mathbf{y})$



Hard positives (green): $S^+ = M(\mathbf{y}) \setminus E(\mathbf{y})$

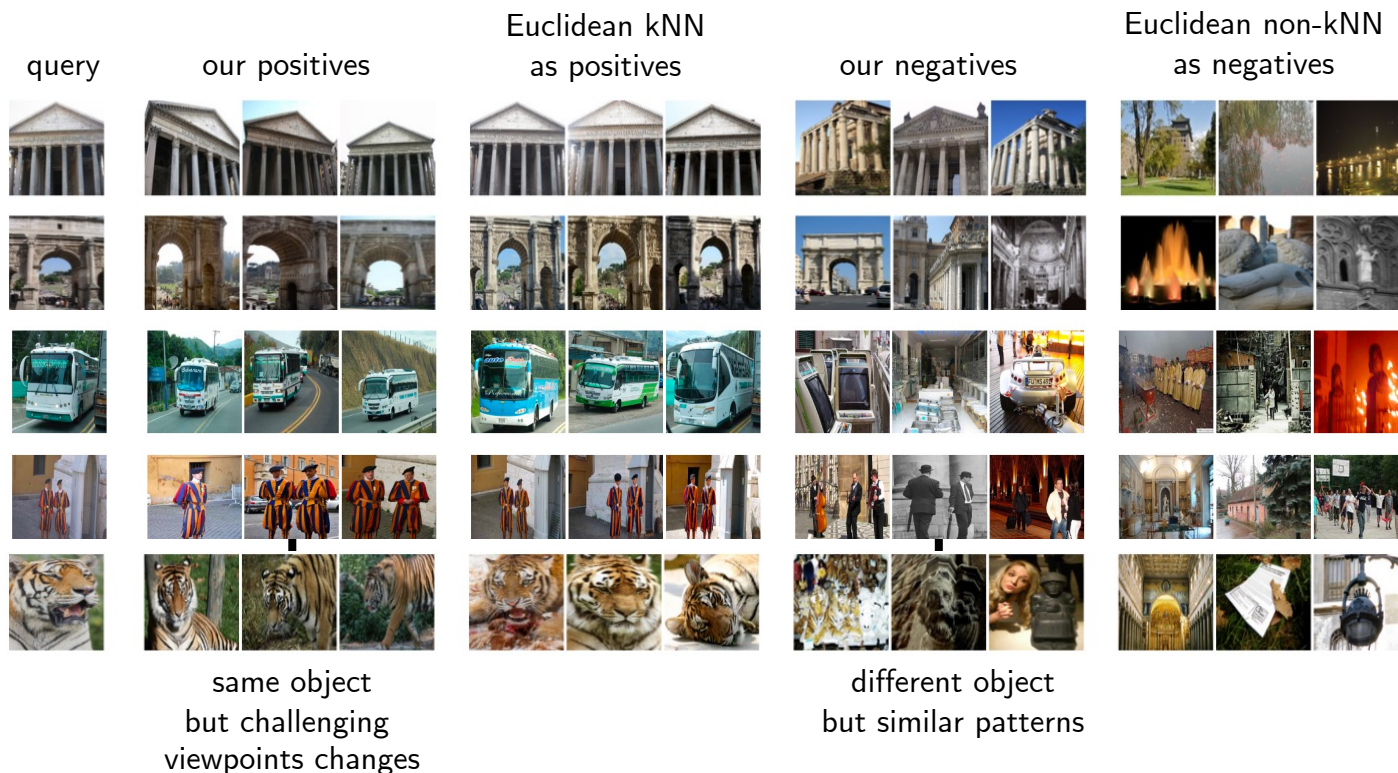


Hard negatives (red): $S^- = E(\mathbf{y}) \setminus M(\mathbf{y})$

● : query

NN = nearest neighbors

MoM: discovery of training pairs from 10^6 images



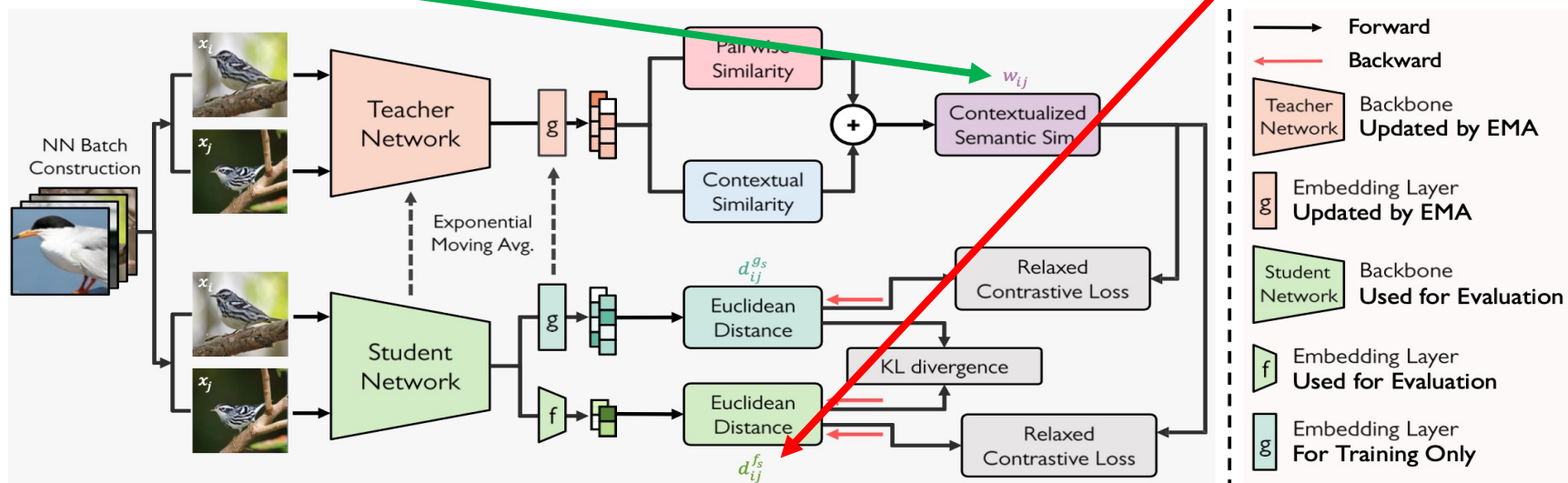
self-taught deep metric learning

- teacher-student scheme with a strong teacher
 - EMA teacher (momentum encoder)
 - larger dimensionality of the teacher representation
 - use of contextual similarity: improved compared to Euclidean
- relaxed contrastive loss: uses soft teacher labels

$$\mathcal{L}_{\text{RC}}(\mathcal{Z}^{f_s}) = \frac{1}{n} \sum_{i=1}^n \sum_{j \neq i}^n \boxed{w_{ij}} \boxed{d_{ij}^{f_s}}^2 + \frac{1}{n} \sum_{i=1}^n \sum_{j \neq i}^n (1 - \boxed{w_{ij}}) [\delta - \boxed{d_{ij}^{f_s}}]^2,$$

soft-labels

distances to optimize



[Kim, Self-Taught Metric Learning without Labels, CVPR2022]

self-taught deep metric learning

Methods	Arch.	CUB	Cars	SOP
		R@1	R@1	R@1
MoCo [20]	G ¹²⁸	48.3	37.2	53.3
BYOL [18]	G ¹²⁸	47.7	31.5	48.7
MSF [31]	G ¹²⁸	46.0	30.5	41.6
STML	G ¹²⁸	59.7	49.0	65.8
MoCo [20]	G ⁵¹²	51.0	39.0	53.4
BYOL [18]	G ⁵¹²	50.9	38.5	51.2
MSF [31]	G ⁵¹²	49.6	32.5	47.8
STML	G ⁵¹²	60.6	50.5	65.3

questions

